Background:

T1 = Garment Vendor: Far Eastern, T2 = Fabric Supplier: Youngtex

Fabric Article#: BIOFLEX-LAB107AW

Material composition: 84% Recycled Nylon, 16% Spandex with Wicking treatment

There are totally 4 colors: Abyss, Bark, Black, Sky Grey.

Phenomenon:

Five dye lots of Bark color, each lot shows noticeable visual shade differences:



DOE Purpose & Evaluation:

The bulk fabric CF test results of BIOFLEX-LAB107AW for Season: SP26.

4 colors bulk fabric samples submitted from T1: Far Eastern to SGS for testing & Test results.

DOE by using CF Testing as Quality Predictive Tool to assess the color tone & shade in noticeable differences.

DOE Findings:

Summarized by the excel file & photo shots as follows -

CF Failures as two problematic colors: 1. Bark & 2. Black

Body Fabric	Style Number	Color	CF to Dye Transfer In Storage Option I: 24 + 3C for 48 hours						
			Color Chang	Acetat	Cotto	Nylor	Polyest	Acryli	Woo
Youngtex - Bioflex Lab - BIOFLEX - LAB107AW	V3035/V3067/V496	Abyss	4.5	4.5	4.5	4.5	4.5	4.5	4.5
			4.5	4.5	4.5	4.5	4.5	4.5	4.5
	V3035/V3067/V496	Bark	4.5	4.5	4.5	4.5	4.5	4.5	4.5
			4.5	4.5	4.5	4	4.5	4.5	4
			4.5	4.5	4.5	4.5	4.5	4.5	4.5
			4.5	4.5	4.5	3.5	4.5	4.5	4.5
			4.5	4	4	3.5	4.5	4.5	4
			4.5	4.5	4.5	4.5	4.5	4.5	4.5
			4	4.5	4	3.5	4.5	4.5	4
	V3035/V3	Black	4.5	4.5	3.5	3	4	4.5	4
	V3067	Sky Grey	4.5	4.5	4.5	4.5	4.5	4.5	4.5

Body Fabric	Style Number	Color	CF to Dye Transfer In Storage Option I: 70C for 48 Hours						
			Color Chang	Acetat	Cotto	Nylor	Polyest	Acryli	Woo
Youngtex - Bioflex Lab - BIOFLEX - LAB107AW	V3035/V3067/V496	Abyss	4.5	4.5	4.5	4.5	4.5	4.5	4.5
			4	4.5	4.5	4	4.5	4.5	4.5
	V3035/V3067/V496	Bark	4.5	4	4	2.5	4.5	4.5	4
			4	4.5	4.5	4	4.5	4.5	4
			3.5	4.5	4.5	2.5	4.5	4.5	4
			3	3.5	3.5	1.5	4	4	2.5
			4	4.5	4.5	3.5	4.5	4.5	4
			3.5	4.5	4	2	4.5	4.5	3.5
			4	4.5	4.5	4	4.5	4.5	4
	V3035/V3	Black	4	4.5	3.5	1.5	4	4	2
	V3067	Sky Grey	4.5	4.5	4.5	4.5	4.5	4.5	4.5

Body Fabric	Style Number	Color	CF to Washing							CF to Light
			Color Chan	Aceta	Cotto	Nylo	Polyes	Acryl	Woo	
Youngtex - Bioflex Lab - BIOFLEX - LAB107AW	V3035/V3067/V496	Abyss	4.5	4.5	4.5	4	4.5	4.5	4.5	4.5
			4.5	4.5	4.5	3.5	4.5	4.5	4.5	4
	V3035/V3067/V496	Bark	4	3.5	3.5	2.5	4	4	4	3.5
			4.5	4	4.5	4	4.5	4.5	4	2.5
			4	4	4.5	2	4.5	4.5	4	4
			4	3.5	3.5	2	3.5	3.5	3.5	3.5
			4	4	4	3.5	4	4.5	3.5	2.5
			4.5	4	4	2	4.5	4.5	3.5	4
			4	4	4	3.5	4	4.5	3.5	2.5
	V3035/V3	Black	4.5	4.5	4.5	2.5	4.5	4.5	4.5	4.5
	V3067	Sky Grey	4	4.5	4.5	4.5	4.5	4.5	4.5	4.5

CF to Perspiration													
Acid							Alkaline						
Color Chang	Aceta	Cotto	Nylo	Polyes	Acryli	Woo	Color Chang	Aceta	Cotto	Nylo	Polyes	Acryli	Woo
4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5
4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5
4.5	4.5	4	4	4.5	4.5	4.5	4.5	4.5	4.5	3	4.5	4.5	4
4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4	4	3.5	4.5	4.5	3.5
4.5	4.5	4.5	4	4.5	4.5	4.5	4.5	4.5	4.5	3	4.5	4.5	4.5
4.5	4.5	4	3.5	4.5	4.5	4.5	4.5	4.5	4	3.5	4.5	4.5	4.5
4.5	4.5	4.5	4	4.5	4.5	4	4.5	4	3.5	3	4	4	3
4.5	4.5	4.5	4	4.5	4.5	4.5	4.5	4.5	4.5	3	4.5	4.5	4
4.5	4.5	4.5	4	4.5	4.5	4	4.5	3.5	3	2.5	3.5	3.5	2.5
4.5	4	4	3	4	4	4	4.5	3.5	3	2.5	3.5	3.5	3.5
4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5

Body Fabric	Style Number	Color	CF to Water						
			Color Chang	Aceta	Cotto	Nylo	Polyes	Acryli	Woo
Youngtex - Bioflex Lab - BIOFLEX - LAB107AW	V3035/V3067/V496	Abyss	4.5	4.5	4.5	4.5	4.5	4.5	4.5
			4.5	4.5	4.5	4.5	4.5	4.5	4.5
	V3035/V3067/V496	Bark	4.5	4.5	4.5	3.5	4.5	4.5	4.5
			4.5	4.5	4.5	4	4.5	4.5	4.5
			4.5	4.5	4.5	4	4.5	4.5	4.5
			4	4.5	4	3	4.5	4.5	4.5
			4.5	4.5	4.5	3.5	4.5	4.5	4
			4.5	4.5	4.5	3.5	4.5	4.5	4.5
			4.5	4.5	4.5	4.5	4.5	4.5	4
	V3035/V3	Black	4.5	4.5	3.5	2	4	4	4
	V3067	Sky Grey	4.5	4.5	4.5	4.5	4.5	4.5	4.5

What Learned from the DOE:

- **ONLY Bark color** resulted in **Color Change (Rating of 3)** under **Dye Transfer in Storage at 70C for 48 hours**.
- This suggests **Bark is more sensitive to elevated temperature at 70C.**
- **CF Failures** in both **Bark color** and **Black color** indicated these **two colors are having Dyeing & Finishing issues such as poor Dyeing, Fixing and Rising.**

Can this DOE be a Quality Predictive tool for Pre-seeing the Potential Color Change or Color Difference Issue?

- **Not reliably**, based on current data - Only one test condition (70C storage) showed a notable result of Color Change (Rating at 3).
- Other colors did not show similar behavior, limiting generalizability.

Do We Know the Root Cause in This Case? Or Any Possible Clues from the DOE?

- **Root cause remains unknown.**

- The only **clue** is the **temperature sensitivity of Bark color**.

Conclusion:

Bark color is **more vulnerable of Color Change (Rating at 3) to high-temperature conditions (70C)**, possibly due to **Acid Dyeing Type & Formulation (Color Recipe).**

CF testing under extreme conditions (70C) **may help flag temperature-sensitive colors (Bark & Black) in this case**, **but** it's **not sufficient** as a standalone predictive tool for pre-seeing the potential color change or shading issues.

What we observed from the DOE**, Bark and Black colors are both failing the normal testing package of CF requirements, meaning the improvement from T2 Fabric supplier – Youngtex is needed.**

Remarks:

Assuming or simulating 70C during the transit (Inside Container Shipment in the summer), our garment is normally packed into the paper bag individually.